3GPP TSG-RAN WG1 Meeting #120 **R1-2500098**
Athens, Greece, February 17th – 21st, 2025

Agenda Item: **9.2.2**
Source: **Huawei, HiSilicon**
Title: **On 128 CSI-RS ports and UE reporting enhancements**
Document for: **Discussion and Decision**

1 Introduction

In this contribution, we provide our views on CSI enhancement for up to 128 CSI-RS ports and UE reporting enhancement for CJT deployments under non-ideal synchronization and backhaul [1].

2 CSI enhancement for up to 128 CSI-RS ports

To increase spectrum efficiency and capacity, it is targeted to support large antenna arrays by enhancement to CSI reporting and hybrid beamforming with up to 128 CSI-RS ports.

2.1 CSI-RS resource configuration

For port indexing of up to 128 ports, the following is agreed.

Agreement
For the Rel-19 Type-I and Type-II codebook refinement for 48, 64, and 128 CSI-RS ports, regarding port mapping,
- Following legacy principle, "sequential ordering/indexing within" a group of Q indices {i_0, i_1, …, i_{Q-1}} is a linearly increasing sequence such that $i_q < i_{q+1}$ (where q=0, 1, …, Q-2 is the port index within a CSI-RS resource, and i_q or $i_{q+1} \in$ {0, 1,…, KQ-1}) is the port index for the codebook, across the K>1 CSI-RS resources).
- After resource aggregation, P (=48, 64, or 128) ports are numbered in accordance to Table 7.4.1.5.3-1 from TS 38.211

Based on the agreement, the port indexing can be expressed as below:

$$p_k = \begin{cases} 3000 + \frac{N}{2}h_k + n_2 v_k + \left\lfloor (s+jL)/n_2 \right\rfloor N_2 + \left((s+jL)\%n_2\right), & s+jL < \frac{N}{2} \\ 3000 + \frac{N}{2}h_k + n_2 v_k + \left\lfloor \left(s+jL-\frac{N}{2}\right)/n_2 \right\rfloor N_2 + \left(\left(s+jL-\frac{N}{2}\right)\%n_2\right) + NK/2, & s+jL \geq \frac{N}{2} \end{cases}$$

where p_k is port index of the ports in the kth CSI-RS resource, $k = 0,\ldots,K-1$. N is the number of ports per CSI-RS resource, and N_2 is the number of rows of the up to 128 port antenna arrays. h_k and v_k are the horizontal and vertical index of kth CSI-RS resource as Figures 1 to 2. And n_1 and n_2 are the number of columns and rows, respectively. The definitions of s, j, L are the same as those in legacy definitions.

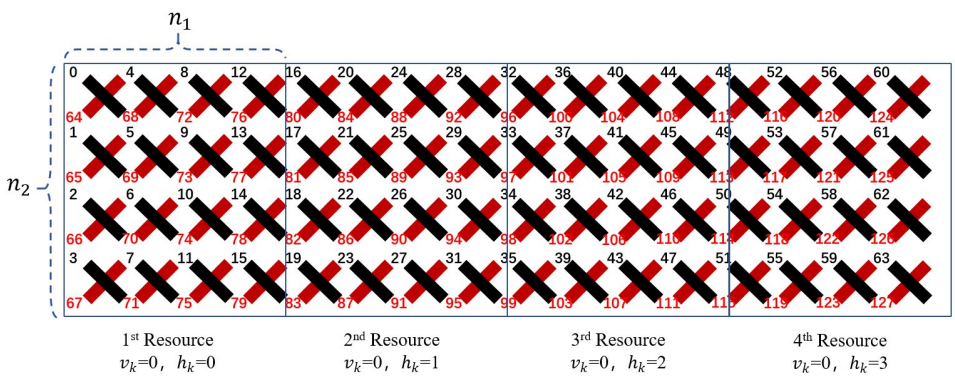


Figure 1 Port numbering for 128-port antenna array with splitting in horizontal dimension

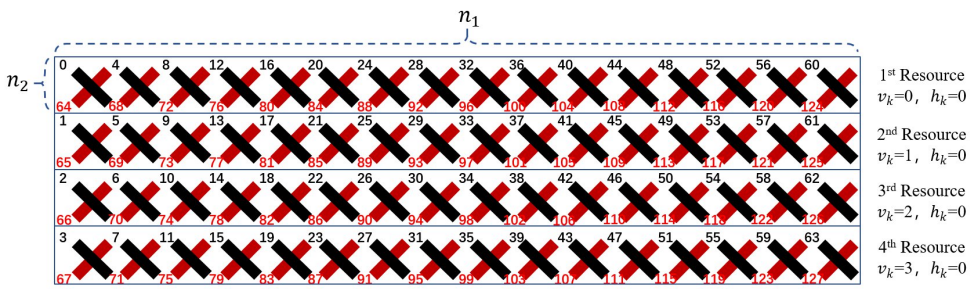


Figure 2 Port numbering for 128-port antenna array with splitting in vertical dimension

Proposal 1: The up to 128 CSI-RS ports are numbered as

$$p_k = \begin{cases} 3000 + \frac{N}{2} h_k + n_2 v_k + \left\lfloor (s + jL)/n_2 \right\rfloor N_2 + \left((s + jL) \% n_2 \right), & s + jL < \frac{N}{2} \\ 3000 + \frac{N}{2} h_k + n_2 v_k + \left\lfloor \left(s + jL - \frac{N}{2} \right) / n_2 \right\rfloor N_2 + \left(\left(s + jL - \frac{N}{2} \right) \% n_2 \right) + NK/2, & s + jL \geq \frac{N}{2} \end{cases}$$

where p_k is port index of the ports in the kth CSI-RS resource, N is the number of ports per CSI-RS resource, N_2 is the number of rows of the antenna panel, h_k and v_k are the horizontal and vertical index of the kth CSI-RS resource, n_1 and n_2 are the number of columns and rows per CSI-RS resource. The definitions of s, j, L are the same as those in legacy definitions.

In Rel-18, Type-I NES is introduced to config N sub-configurations of port shutdown patterns by bitmaps of up to 32 bits, for up to 32 CSI-RS ports. Then UE reports the N CSI(s) according to the shutdown patterns of N sub-configurations to assist the gNB in energy saving. In RAN1#118 meeting, to support up to 128 ports, the length of bitmap is increased to 48, 64 and 128, for 48, 64 and 128 CSI-RS ports. As a result, the total overhead escalates from $N \times 32$ to $N \times 128$, where N is the number of Type-I sub-configuration settings configured by the gNB. This increases the RRC overhead significantly.

Considering the antenna patterns are always rectangular array with cross polarization, the pattern of NES sub-configurations can be indicated by a bitmap in N1-dimension and a bitmap in N2-dimension to substantially reduce the overhead for large-port Type-I NES configurations. One example is illustrated in Figure 3, where a sub-configuration of 8 ports is configured for 32 CSI-RS ports in total. The configuration for the N1-dimension is [01111000], and [1 0] for the N2-dimension. For 128-port with (N1, N2)=(16, 4), the #bits per sub-configuration is reduced from 128 to 20, resulting in a savings of $N \times 108$ bits overall.

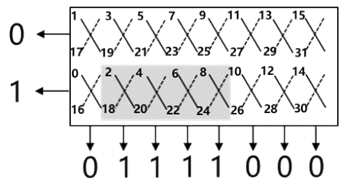


Figure 3 Two-level bitmap

Proposal 2: Support to reduce the overhead of RRC configuration for Rel-19 Type-I NES.

2.2 Enhancements for Type-I codebook

2.2.1 UE feature of Type-I SP codebook scheme

As agreed in RAN1 #117, Scheme A and Scheme B are introduced for Type-I codebook enhancement in Rel-19 to support both RI=1-4 and RI=5-8. However, the RI=1-4 and RI=5-8 for the same scheme also have different performance and implementation complexity.

For Scheme-A, the RI=1-4 extends the (N1, N2) values of legacy Type-I codebook. There is limitation on the selection of the second beam, which reduces the implementation complexity. However, this also potentially degrades performance when more beams are available for larger than 32 ports. Meanwhile, for RI 5-8, the second, third and possible fourth beams are selected freely as long as they are orthogonal in at least one dimension. This increases the number of candidates precoders thus increases complexity. At the same time, the performance can be improved.

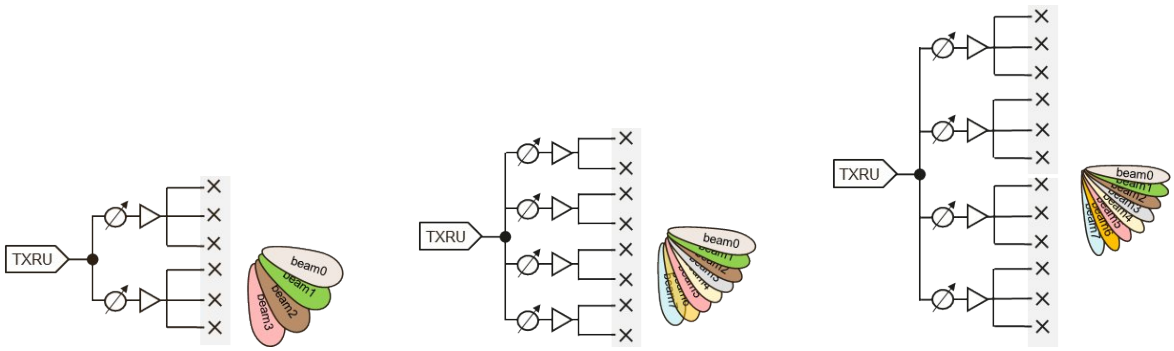
For Scheme-B RI=1-4, UE can select the beams and also the inter-polarization co-phasing freely for each layer, which improves performance while increases number of candidates precoders thus the implementation complexity. Meanwhile for Scheme-B RI=5-8, the beams can only be selected from N1*N2 orthogonal beams assuming a common offset (q1, q2). And the inter-polarization co-phasing is also limited for layers sharing the same beam. This can reduce the number of candidate precoders thus reduce the implementation complexity, however, the performance is also degraded potentially.

Based on the above analysis, the Scheme-B RI=1-4 has better performance than Scheme-A RI=1-4 with more complexity, while Scheme-A RI=5-8 has better performance than Scheme-B RI=5-8 with more complexity. For performance targeted UEs or cost targeted UEs, it should be allowed to support both Scheme-B RI=1-4 and Scheme-A RI=5-8, or Scheme-A RI=1-4 and Scheme-B RI=5-8.

Proposal 3: UE can report to support both Scheme-B RI=1-4 and Scheme-A RI=5-8, or both Scheme-A RI=1-4 and Scheme-B RI=5-8.

2.3 Multi-beam reporting for hybrid beamforming

As is known to all, larger antenna array with hybrid beamforming (HBF) can achieve reasonable tradeoff between cost-effectiveness and performance in FR1. The typical HBF subarray architectures are exemplified in Figure 4. With the increase of subarray size, the analog beam becomes narrower, which means more analog beams are needed to guarantee the coverage. Specifically, the number of required analog beams for the typical subarray architecture in Figure 4 is 2/4 (corresponding to the number of phase shifters) without oversampling and can reach 8 under double oversampling. To accommodate the multi-beam based HBF transmission, the corresponding CSI measurement and reporting are discussed in the following sections.



(a) Subarray size 6 **(b) Subarray size 8** **(c) Subarray size 12**

Figure 4 Typical HBF subarray architectures

2.3.1 Multi-beam channel measurement

In RAN1 #116 meeting, the supported combinations of the number of CSI-RS resources K_s and the maximum number of ports per resource were agreed for multi-beam channel measurement as below [4]. Following are the approaches confronting the large CSI-RS and configuration overhead incurred by multi-beam channel measurement.

Agreement

For the Rel-19 CRI-based CSI refinement for up to 128 CSI-RS ports, the supported combinations of K_S value and the maximum number of ports per NZP CSI-RS resource are as follows:

- FFS: UE capability on K_S and the number of ports per resource

K_S	Maximum # ports per resource
2, 3, 4	32
5, 6, 7, 8	16

2.3.1.1 CSI-RS overhead reduction

Intuitively, each analog beam can be associated with a separate CSI-RS resource during multi-beam channel measurement, which leads to the linearly-increased CSI-RS overhead shown in Table 1. However, if oversampling, the conventional method to improve the coverage, is adopted to generate the analog beams, transmitting less CSI-RS resources than the total number of analog beams without harming CSI measurement is

possible. This attributes to the wireless channel sharing among analog beams, which makes it feasible to reconstruct the channel of oversampled beams through that of orthogonal beams at the UE side.

Table 1 CSI-RS overhead for multi-beam channel measurement (CSI-RS periodicity 10ms, 16 ports)

Total number of analog beams	CSI-RS overhead based on one-one mapping between analog beams and CSI-RS resources	CSI-RS overhead based on mapping only between orthogonal beams and CSI-RS resources
8	7.7%	3.8%

Observation 1: For multi-beam channel measurement, the CSI-RS overhead is pretty high if each analog beam is associated with a separate CSI-RS resource.

Proposal 4: For multi-beam channel measurement, transmitting less CSI-RS resources than the total number of analog beams should be considered to reduce the CSI-RS overhead.

2.3.1.2 Configuration overhead reduction

For legacy aperiodic CSI-RS, a resource-set-level *aperiodicTriggeringOffset* representing *Offset X between the slot containing the DCI that triggers a set of aperiodic NZP CSI-RS resources and the slot in which the CSI-RS resource set is transmitted* is configured. However, given that gNB transmits CSI-RS resources corresponding to different analog beams in TDM manner under HBF architecture, which means multi-beam channel measurement can distribute among up to K_s slots, the legacy aperiodic configuration is regrettably incompetent.

Facing above dilemma, a straightforward solution is to adopt two-level offset (i.e., reusing legacy resource-set-level *aperiodicTriggeringOffset* and introducing additive resource-level offset), as agreed in last meeting [119].

Agreement

For the Rel-19 CRI-based CSI refinement for up to 128 CSI-RS ports, regarding aperiodic CSI-RS resource configuration, an RRC-configured resource-level slot offset (relative to the resource-set-level slot offset, using the same design as Rel-19 Type-I/II codebook refinement for 48, 64, and 128 ports) is supported for aperiodic CSI-RS resource set.

- FFS: The number of bits for indicating the resource-level slot offset (relative to the resource-set-level slot offset) for K_S resources, including the value(s) of the slot offset.
- FFS: Whether, in addition, configuring an *available* slot offset for each CSI-RS resource within the aperiodic CSI-RS resource set.
    - Note: "*Available* slot offset" is analogous to the Rel-17 SRS triggering offset enhancement.

Nevertheless, considering that multi-beam channel measurement distributing among up to K_s slots can experience one or multiple times UL/DL switching, the agreed two-level offset is unfortunately inefficient. Specifically, as shown in Figure 5, when the DCI is transmitted in different slot, imputing to the existence of UL/DL switching, configuring different *aperiodicTriggeringOffset* while reusing the same set of CSI-RS resource configuration becomes an unrealistic notion.

In order to overcome above inefficiency, the design principle of available slot offset introduced in Rel-17 SRS can be reused. Concretely, as depicted in Figure 6, no matter which slot the DCI is transmitted in, through the perfect cooperation between resource-set-level *aperiodicTriggeringOffset* and resource-level available slot offset, a single set of CSI-RS resource configuration is adequate, which significantly saves configuration overhead and avoids meaningless waste of UE capability.

Observation 2: The design principle of available slot offset introduced in Rel-17 SRS should be reused for the aperiodic CSI-RS resource configuration of multi-beam channel measurement.

Furthermore, due to the wide span of multi-beam channel measurement, the value range of available slot offset is relatively large. Taking the configuration overhead and forward compatibility (i.e., in case more beams need to be measured in the future) into account, available slot differential offset as illustrated in Figure 7 should be supported.

Proposal 5: Support available slot differential offset for the aperiodic CSI-RS resource configuration of multi-beam channel measurement.

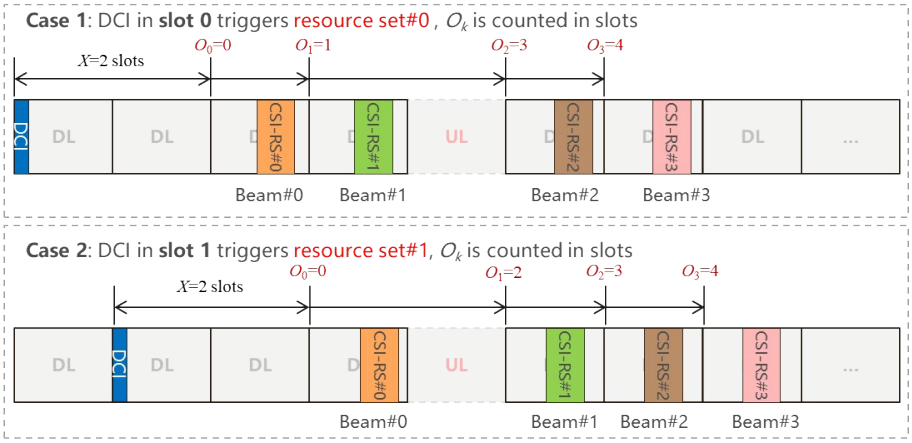


Figure 5 Two-level offset for aperiodic multi-beam channel measurement

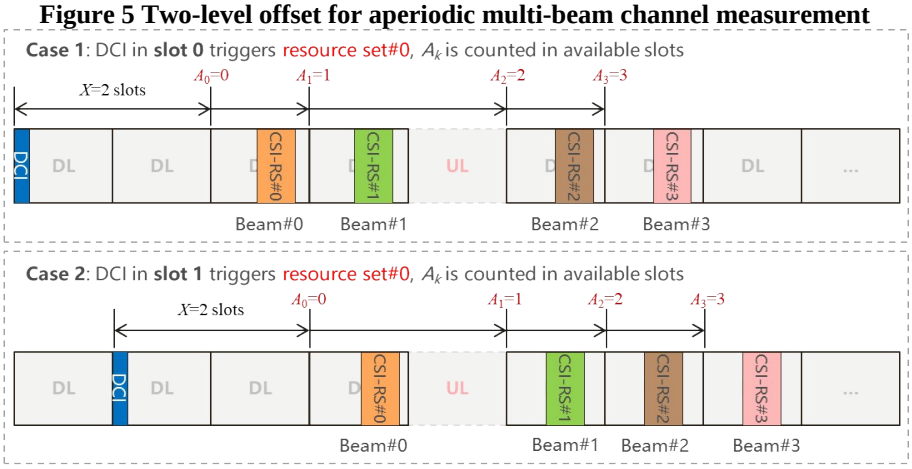


Figure 6 Available slot offset for aperiodic multi-beam channel measurement

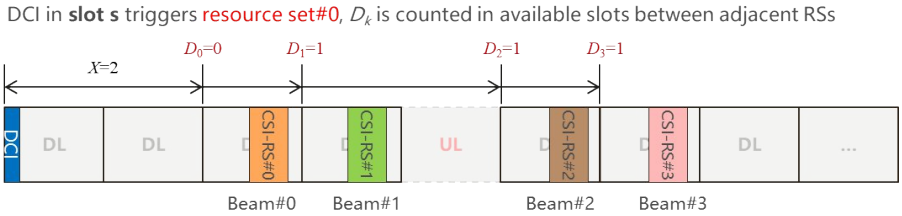


Figure 7 Available slot differential offset for aperiodic multi-beam channel measurement

2.3.2 Multi-beam CSI reporting

Regarding multi-beam CSI reporting, several key agreements were achieved in RAN1 #118 ~ RAN1 #119 [1]. The remaining issue listed below is subsequently discussed:

- UCI design for multiple CCs

2.3.2.1 UCI design for multiple CCs

Considering that the gNB can only transmit with different analog beams in TDM manner under HBF architecture, when it comes to intra-band CA scenario, the gNB may not be able to schedule all CCs in a band for UE if per-CC independent CSI reporting is applied. As shown in Figure 8, assuming a UE reports Beam #0 and Beam #1 for CC #0 and #1, while reports Beam #2 and Beam #3 for CC #2 and #3, subject to the incomplete CSI, by no means can gNB simultaneously schedule all CCs for the UE with a certain beam, which incurs spectrum efficiency degradation. In order to avoid similar situation, the UE should report same beams for multiple CCs in a band.

Observation 3: Under intra-band CA scenario, per-CC independent CSI reporting may lead to spectrum efficiency degradation.

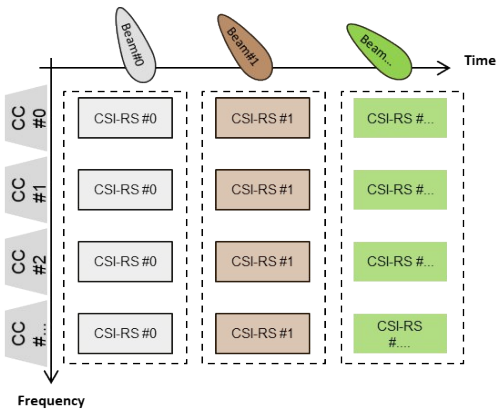


Figure 8 Multi-beam scheduling over multiple CCs

3 UE reporting enhancement for CJT

3.1 Linkage between DO report and CJT CSI reporting

Agreement
For the Rel-19 aperiodic standalone CJT calibration (CJTC) reporting, when linking CJTC Dd and Rel-18 eType-II CJT CSI reports is configured with a joint trigger, the timeline (Z/Z') is determined as Z/Z' associated with the Rel-18 eType-II CJT, plus D_{relax}

- The value of D_{relax} is a UE capability, taken from {0, d_{relax}}
  - FFS: The value of d_{relax} (>0), including whether it depends on SCS
- For linking CJTC Dd and Rel-18 eType-II CJT CSI, joint triggering is a separate UE feature group from separate triggering

In the case when CJTC Dd and Rel-18 eType-II CJT CSI reports are configured with a joint trigger, UE may need extra processing timeline relaxation d_{relax} based on the Z/Z' associated with Rel-18 eType-II CJT. One open issue is whether the extra relaxation depends on SCS. Since the Rel-18 eType-II CJT Z/Z' has considered the SCS, there is no need to further consider it in D_{relax}. The d_{relax} can be a fixed number of symbols, e.g., 14 or 28 symbols.

Proposal 6: Support d_{relax} as a fixed number of symbols, e.g., 14 or 28 symbols.

3.2 Phase offset reporting

Agreement
For the Rel-19 aperiodic standalone CJT calibration (CJTC) reporting, when *ReportQuantity* is *'cjtc-P'* (DL/UL phase offset), the UE assumes that the N_{TRP} CSI-RS resources are transmitted without DL/UL switching in between the N_{TRP} resources

One open issue of PO reporting is how to define the subband grid in frequency domain (i.e. the start position for each subband). Considering that the related CJT CSI feedback (PMI) is based on CSI subband, if the grids of the PO reporting subband and the CSI reporting subband are not aligned, there will be a large jump for CSI calculation at the boundary of PO subbands, which degrades the accuracy of CSI feedback. Moreover, aligning the PO reporting subband for different UEs is also beneficial for gNB to achieve joint optimal PO compensation for the network (thus for multiple paired Ues). Therefore, one straightforward way is to align the boundary of the PO subband with that of the existing CSI reporting subband. When the configured sub-band size for PO reporting ({1, 2, 4, 8, 16} PRB) is different from the configured CSI reporting subband size ({4, 8, 16, 32} PRB), one CSI reporting subband could be evenly divided into multiple PO reporting subbands or multiple CSI subbands could be combined into one PO reporting subband.

Proposal 7: For subband reporting of PO, the boundary of the PO subband should be aligned with that of the existing CSI reporting subband .

4 Conclusions

The contribution provides our considerations on 128 CSI-RS ports and UE reporting enhancements for CJT, and the observations are listed as following:

Observation 1: For multi-beam channel measurement, the CSI-RS overhead is pretty high if each analog beam is associated with a separate CSI-RS resource.

Observation 2: The design principle of available slot offset introduced in Rel-17 SRS should be reused for the aperiodic CSI-RS resource configuration of multi-beam channel measurement.

Observation 3: Under intra-band CA scenario, per-CC independent CSI reporting may lead to spectrum efficiency degradation.

The proposals are listed as below:

Proposal 1: The up to 128 CSI-RS ports are numbered as

$$p_k = \begin{cases} 3000 + \frac{N}{2} h_k + n_2 v_k + \left\lfloor (s + jL)/n_2 \right\rfloor N_2 + \left((s + jL) \% n_2 \right), & s + jL < \frac{N}{2} \\ 3000 + \frac{N}{2} h_k + n_2 v_k + \left\lfloor \left(s + jL - \frac{N}{2} \right) / n_2 \right\rfloor N_2 + \left(\left(s + jL - \frac{N}{2} \right) \% n_2 \right) + NK/2, & s + jL \geq \frac{N}{2} \end{cases}$$

where p_k is port index of the ports in the kth CSI-RS resource, N is the number of ports per CSI-RS resource, N_2 is the number of rows of the antenna panel, h_k and v_k are the horizontal and vertical index of the kth CSI-RS resource, n_1 and n_2 are the number of columns and rows per CSI-RS resource. The definitions of s, j, L are the same as those in legacy definitions.

Proposal 2: Support to reduce the overhead of RRC configuration for Rel-19 Type-I NES.

Proposal 3: UE can report to support both Scheme-B RI=1-4 and Scheme-A RI=5-8, or both Scheme-A RI=1-4 and Scheme-B RI=5-8.

Proposal 4: For multi-beam channel measurement, transmitting less CSI-RS resources than the total number of analog beams should be considered to reduce the CSI-RS overhead.

Proposal 5: Support available slot differential offset for the aperiodic CSI-RS resource configuration of multi-beam channel measurement.

Proposal 6: Support d_{relax} as a fixed number of symbols, e.g., 14 or 28 symbols.

Proposal 7: For subband reporting of PO, the boundary of the PO subband should be aligned with that of the existing CSI reporting subband .

References

[1] Report of RAN1#119, November 18th – 22nd, 2024.

[2] Report of RAN1#118, August 19th – 23rd, 2024.

[3] Report of RAN1#117, May 20 – 24, 2024.

[4] Report of RAN1#116, Feb. 26 – Mar. 1, 2024.

Appendix A: System simulation assumptions for CSI enhancement for up to 128 CSI-RS ports

Table A-1. Evaluation assumptions for CSI enhancement for up to 128 CSI-RS ports

Parameters	Evaluation assumptions
Layout	Hexagonal grid, 3 sectors per site, 7 macro sites
Scenario	Dense Urban (Macro only)
Carrier frequency	FR1 only, 2.1GHz, with duplexing gap of 200MHz
Subcarrier spacing	15kHz
Simulation Bandwidth	10MHz
BS Tx Power	41 dBm

BS antenna configuration	128 ports: (M, N, P, Mg, Ng; Mp, Np) = (12,16,2,1,1,4,16), (dH,dV) = (0.5, 0.8)λ 64 ports: (M, N, P, Mg, Ng; Mp, Np) = (12,8,2,1,1,4,8), (dH,dV) = (0.5, 0.8)λ 32 ports: (M, N, P, Mg, Ng; Mp, Np) = (8,8,2,1,1,2,8), (dH,dV) = (0.5, 0.8)λ
UE antenna configuration	(M, N, P, Mg, Ng) = (1,2,2,1,1);
UE distribution	80% indoor, 3km/h; 20% outdoor, 30km/h;
UE receiver	MMSE-IRC
MIMO mode	MU-MIMO, rank adaptation
CSI feedback	Feedback assumption: • CSI feedback periodicity: 5ms • Scheduling delay: 4 ms
SRS Configuration	• SRS periodicity with 10ms • Comb: 2 • Number of OFDM symbols: 2
Traffic model	FTP model 1 with packet size 0.5 Mbytes
Traffic load	70% RU

Appendix B: System simulation assumptions for Hybrid Beamforming

Table B-1 System simulation parameters for massive MIMO with hybrid beamforming

Parameter	Value
Duplex, Waveform	TDD, OFDM
Carrier Frequency	7 GHz
Channel Model	According to the TR 38.901
Scenario	UMa with 300 m ISD
Antenna setup and port layouts at gNB	(M, N, P, Mg, Ng; Mp, Np) = (32, 16, 2, 1, 1, 4, 16). (dH, dV) = (0.5, 0.8) 4 analog beams
Antenna setup and port layouts at UE	(M, N, P, Mg, Ng; Mp, Np) = (1,2,2,1,1,1,2), (dH, dV) = (0.5, 0.5)
BS Tx power	46 dBm
BS antenna height	25 m
UE receiver noise figure	9 dB
MIMO scheme	SU/MU-MIMO with rank adaptation Maximum rank = 2 per UE
Traffic model	Full buffer
UE distribution	80% indoor (3km/h), 20% outdoor (30km/h)
UE receiver	MMSE-IRC
Network Layout	7*3 cell, 30 UEs pre cell
Precoding granularity	4RB
Precoding method	EZF

Appendix C: Link level simulation parameters for UE reporting enhancement for CJT

Table C-1 Simulation assumptions of LLS for UE reporting enhancement for CJT

Parameter	Value
Duplex, Waveform	TDD, OFDM
Carrier Frequency	3.5 GHz

Channel Model	CDL-C channel model in TR 38.901
Delay Spread	300ns
BS antenna configuration	(M, N, P, Mg, Ng; Mp, Np) = (2, 8, 2, 1, 1; 2, 8), (dH, dV) = (0.5, 0.8)
TRP number	2
UE antenna configuration	(M, N, P, Mg, Ng; Mp, Np) = (1, 2, 2, 1, 1; 1, 2), (dH, dV) = (0.5, 0.5)
UE number	4
MCS	Link Adaption
Bandwidth	20RB
Numerology	14 OFDM symbol per slot, 30kHz SCS
MIMO Rank	rank = 2 per UE
UE speed	30km/h
Precoding granularity	2RB
DMRS	Type 2 DMRS, double-symbol
DL DMRS channel estimation	LMMSE channel estimation
Periodicity of DO reports	100ms
UCI BLER	0.1%